

Module 12

Assignment 12A: Integrating OpenAI

Laravel + OpenAI API (gpt-4o-mini)

Course: CS 85 - PHP Programming

Instructor: Vicky Seno

Repository: github.com/SergeHall/cs85-php-programming/tree/main/assignments/module12a

OpenAI quickstart: platform.openai.com/docs/quickstart

Outcome

Assignment 12A is connected to the live OpenAI API. Laravel validates the form, builds an adaptive prompt, calls gpt-4o-mini, and returns an editable draft while the API key remains on the server.

Implementation objective. Build a focused content generator that supports a blog post, meta description, or email subject line with professional, casual, and humorous tones.

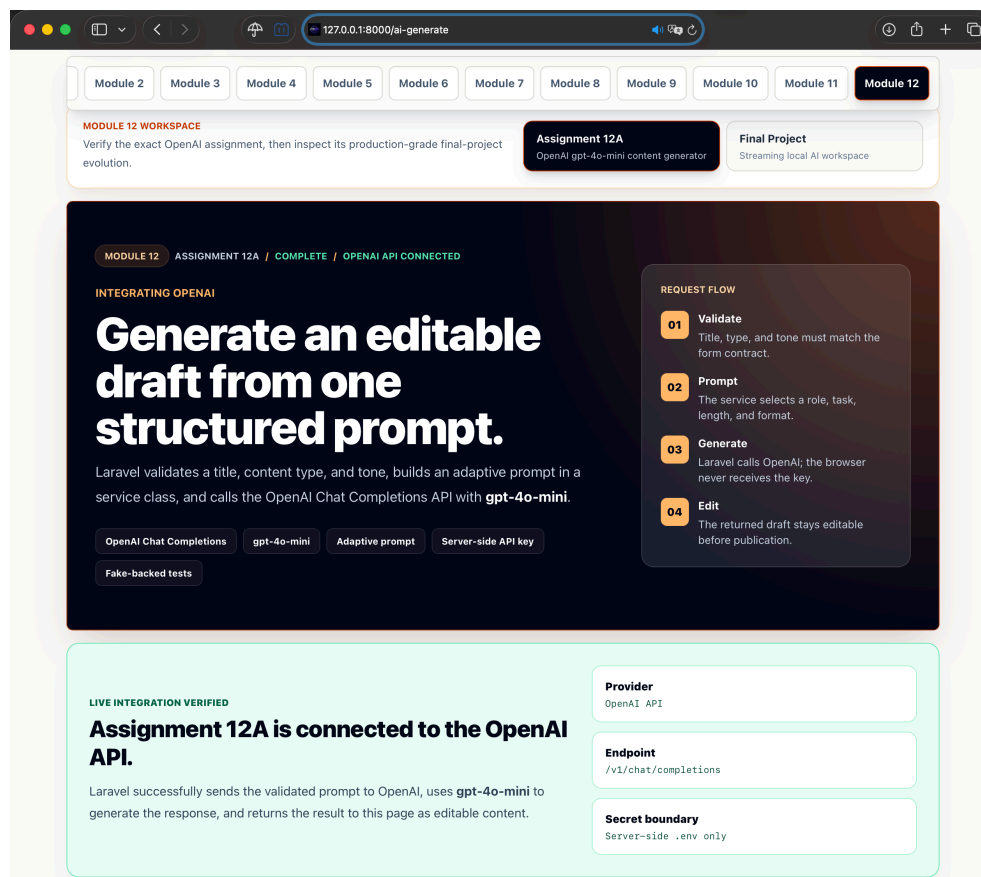
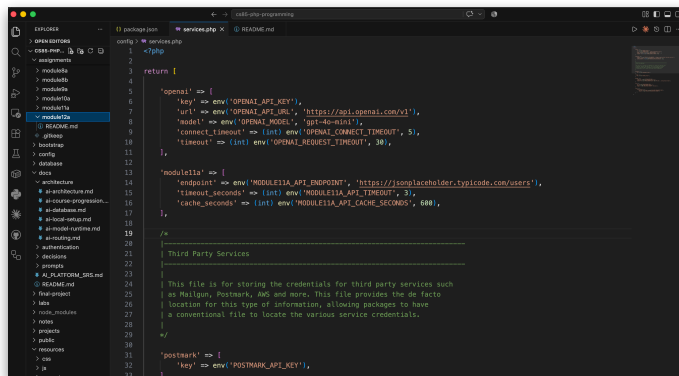


Figure 1. Assignment 12A page and verified OpenAI API boundary

1. How the OpenAI API Was Connected

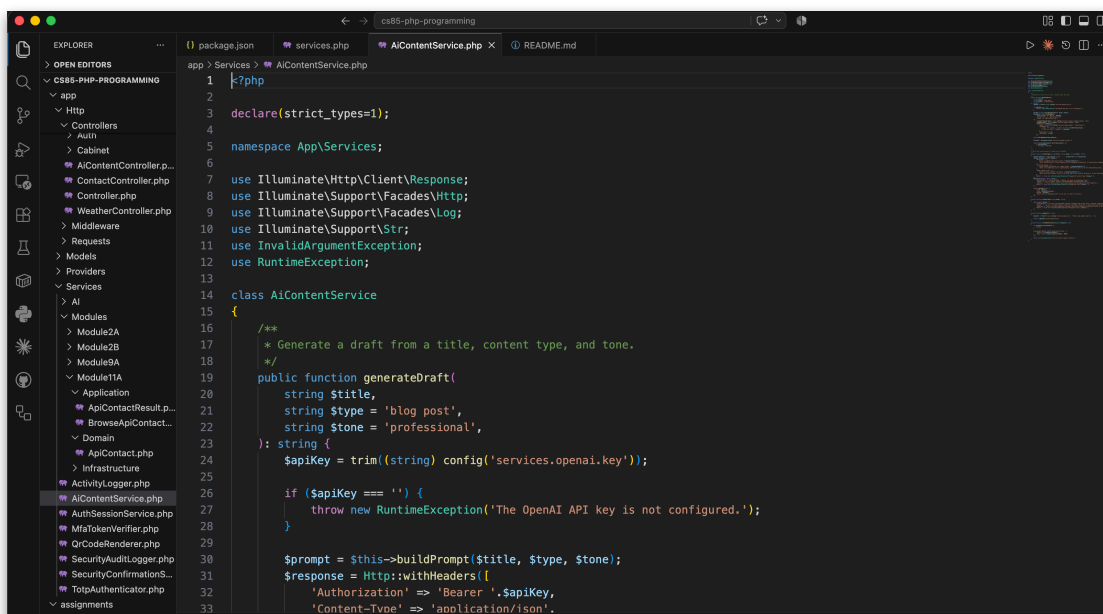
The integration follows a server-side boundary. The browser never receives the secret. Laravel reads configuration from the environment, prepares the prompt, and sends the authenticated request to OpenAI.

CODE 1 | .env and config/services.php (secret redacted)



```
1 <?php
2
3 return [
4
5     'openai' => [
6         'key' => env('OPENAI_API_KEY'),
7         'url' => env('OPENAI_API_URL', 'https://api.openai.com/v1'),
8         'model' => env('OPENAI_MODEL', 'gpt-4o-mini'),
9         'connect_timeout' => (int) env('OPENAI_CONNECT_TIMEOUT', 5),
10        'timeout' => (int) env('OPENAI_REQUEST_TIMEOUT', 30),
11    ],
12
13    'modulellm' => [
14        'endpoint' => env('MODULELLM_API_ENDPOINT', 'https://jsonplaceholder.typicode.com/users'),
15        'timeout_seconds' => (int) env('MODULELLM_API_TIMEOUT', 3),
16        'cache_seconds' => (int) env('MODULELLM_API_CACHE_SECONDS', 600),
17    ],
18
19 ];
20
21 // Third Party Services
22
23
24 // This file is for storing the credentials for third party services such
25 // as Mailgun, Postmark, AWS and more. This file provides the de facto
26 // location for this type of information, allowing packages to have
27 // a conventional file to locate the various service credentials.
```

CODE 2 | AiContentService sends the Chat Completions request



```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace App\Services;
6
7 use Illuminate\Http\Client\Response;
8 use Illuminate\Support\Facades\Http;
9 use Illuminate\Support\Facades\Log;
10 use Illuminate\Support\Str;
11 use InvalidArgumentException;
12 use RuntimeException;
13
14 class AiContentService
15 {
16     /**
17      * Generate a draft from a title, content type, and tone.
18      */
19     public function generateDraft(
20         string $title,
21         string $type = 'blog post',
22         string $tone = 'professional',
23     ): string {
24         $apiKey = trim((string) config('services.openai.key'));
25
26         if ($apiKey === '') {
27             throw new RuntimeException('The OpenAI API key is not configured.');
```

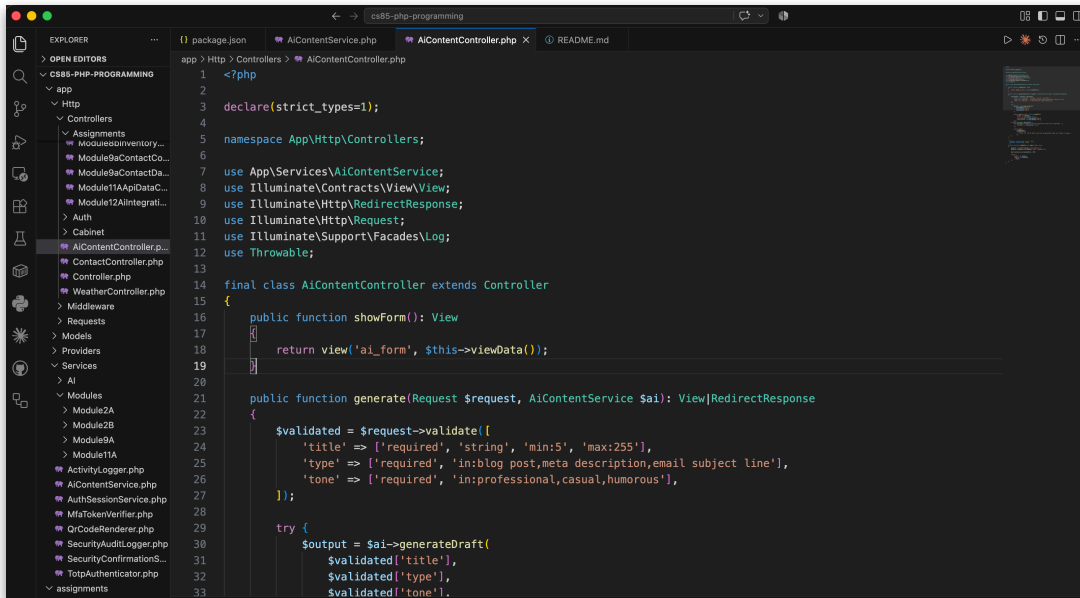
Request flow

Step	Owner	What happens
01	Blade form	Posts title, content type, tone, and CSRF token to Laravel.
02	Controller	Validates the allowlisted values and delegates to AiContentService.
03	Service	Builds the adaptive prompt and calls OpenAI with the server-side key.
04	Blade result	Receives assistant content and keeps it editable before publishing.

Security boundary. The real key is excluded from Git through .gitignore, is not printed in logs or screenshots, and is used only by the Laravel HTTP client.

2. Laravel Coordination and Browser Result

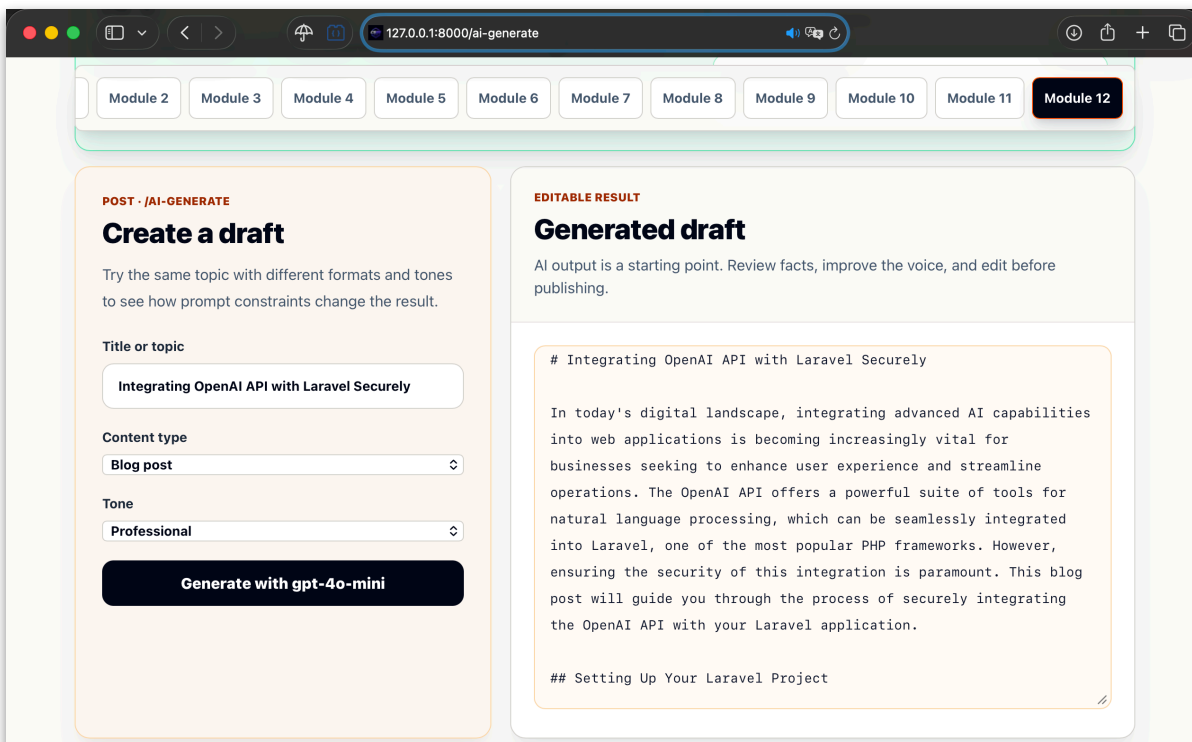
CODE 3 | Routes and controller validation



```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace App\Http\Controllers;
6
7 use App\Services\AiContentService;
8 use Illuminate\Contracts\View\View;
9 use Illuminate\Http\RedirectResponse;
10 use Illuminate\Http\Request;
11 use Illuminate\Support\Facades\Log;
12 use Throwable;
13
14 final class AiContentController extends Controller
15 {
16     public function showForm(): View
17     {
18         return view('ai_form', $this->viewData());
19     }
20
21     public function generate(Request $request, AiContentService $ai): View\RedirectResponse
22     {
23         $validated = $request->validate([
24             'title' => ['required', 'string', 'min:5', 'max:255'],
25             'type' => ['required', 'in:blog post,meta description,email subject line'],
26             'tone' => ['required', 'in:professional,casual,humorous'],
27         ]);
28
29         try {
30             $output = $ai->generateDraft(
31                 $validated['title'],
32                 $validated['type'],
33                 $validated['tone']
34             );
35         } catch (Throwable $e) {
36             Log::error($e->getMessage());
37             return view('ai_form', $this->viewData());
38         }
39
40         return view('ai_form', $this->viewData());
41     }
42 }
```

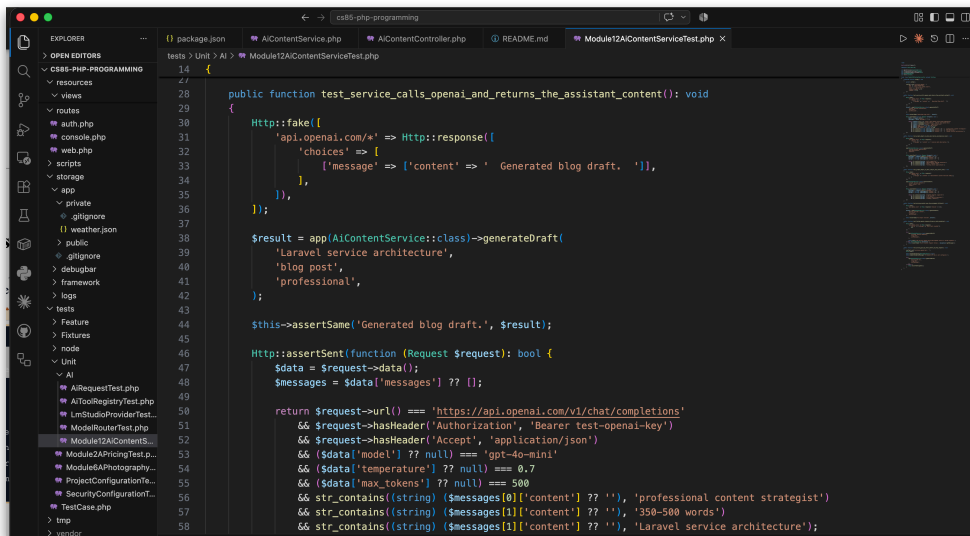
Adaptive prompting. AiContentService selects a role from the requested tone and a format constraint from the requested content type. This gives one model three focused writing modes without duplicating controller logic.

Figure 2. Live gpt-4o-mini response displayed as editable content



3. Testing, Run Instructions, and Evaluation

Automated tests do not spend API credits. Laravel HTTP fakes verify the exact endpoint, Authorization header, model, prompt roles, response extraction, and failure behavior.



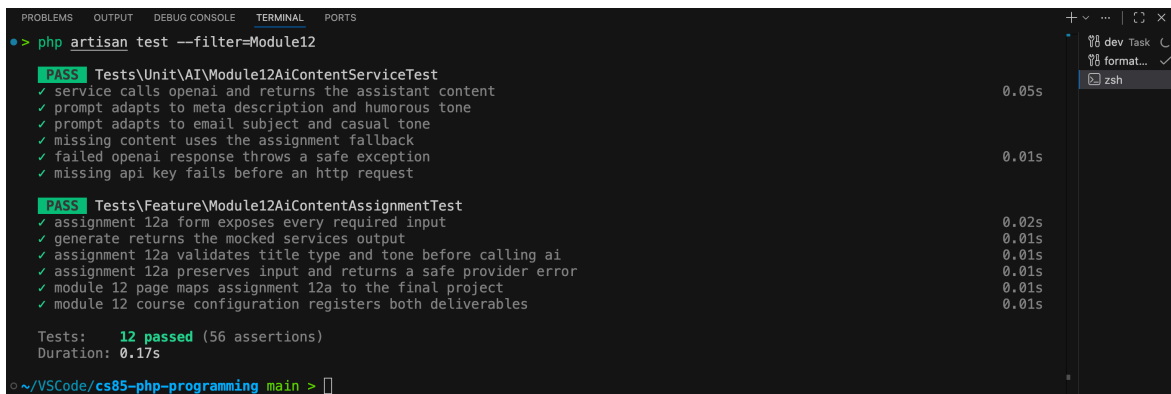
CODE 4 | Fake-backed OpenAI contract test

Verification summary

PASS	Full Laravel suite 204 tests passed with 1,677 assertions.
PASS	Module 12 tests Service, prompt variants, validation, and safe failures pass.
PASS	Code style Laravel Pint completed successfully.
PASS	Frontend build Vite production build completed successfully.
PASS	Live API smoke test OpenAI returned editable content through POST /ai-generate.

Run and test

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Conclusion

The assignment meets the required OpenAI integration goal and also adds a clean service boundary, adaptive prompts, timeout handling, safe provider errors, and fake-backed tests. The implementation is intentionally separate from the larger Final Project AI workspace, making the Module 12A rubric easy to review.